

Tags: Probability

Concentration Inequalities

We bound the probability of a random variable being far from the expectation.

Theorem 1 (Markov Inequality). *Let $X \geq 0$ be a random variable. Then, for any $a > 0$, $\Pr[X \geq a] \leq \mathbb{E}[X]/a$.*

Idea. We split the summation/integral into two parts $x < a$ and $x \geq a$.

Proof. Case (I) Discrete.

Let $X \geq 0$ be discrete random variable taking values in $\{x_i\}$ and $p(x)$ be probability mass function of X .

$$\mathbb{E}[X] = \sum_{x_i} x_i p(x_i) = \sum_{x_i < a} x_i p(x_i) + \sum_{x_i \geq a} x_i p(x_i) \leq \sum_{x_i \geq a} a p(x_i) = a \Pr[X \geq a]$$

By rearranging, we get the inequality. □

Case (II) Continuous.

Let $X \geq 0$ be continuous random variable and $f(x)$ be probability density function of X .

$$\mathbb{E}[X] = \int_0^\infty x f(x) dx = \int_0^a x f(x) dx + \int_a^\infty x f(x) dx \leq \int_a^\infty a f(x) dx = a \Pr[X \geq a]$$

By rearranging, we get the inequality. □

Note that Markov is a first order inequality. We can use Markov inequality to get similar higher order inequalities (higher order inequalities give tighter bounds on probability). Particularly, we apply Markov inequality to $\phi(|X - \mathbb{E}[X]|)$ where $\phi(t)$ is an increasing function in t .

Theorem 2 (Chebyshev Inequality). *Let X be a random variable. Then, for any $a > 0$, $\Pr[|X - \mathbb{E}[X]| \geq a] \leq \text{Var}[X]/a^2$*

Proof. Let X be a random variable with $\mathbb{E}[X] = \mu$. Let $\phi(t) = t^2$. Using Markov inequality,

$$\begin{aligned} \Pr[|X - \mu| \geq a] &= \Pr[\phi(|X - \mu|) \geq \phi(a)] \leq \mathbb{E}[\phi(|X - \mu|)]/\phi(a) \\ &\leq \mathbb{E}[|X - \mu|^2]/a^2 = \text{Var}[X]/a^2 \end{aligned}$$
□

As mentioned earlier, with different ϕ we can get different higher order inequalities. Using $\phi(t) = e^{\lambda t}$ gives us Chernoff inequality. Since, $e^x = 1 + x/1! + x^2/2! + x^3/3! \dots$, Chernoff captures inequalities for every order. This is the same reason why Gaussian kernel is used in SVM.

Theorem 3 (Chernoff Inequality). *Let X be a random variable. Then, $\Pr[X \geq t] \leq e^{-\psi_X^*(t)}$ for $t > 0$, where ψ_X^* is the Cramer transform of X (See Proof).*

Proof. Let X be a random variable and $a > 0$. Using Markov inequality and $\phi(t) = e^{\lambda t}$,

$$\begin{aligned} \Pr[X \geq a] &= \Pr[\phi(X) \geq \phi(a)] \leq \mathbb{E}[\phi(X)]/\phi(a) \\ &\leq \mathbb{E}[e^{\lambda X}]/e^{\lambda a} = \exp(-(\lambda a - \ln(\mathbb{E}[e^{\lambda X}])) \end{aligned}$$

Let $\psi_X(\lambda) = \ln(\mathbb{E}[e^{\lambda X}])$ (ψ_X is referred as cumulant function of X and $\mathbb{E}[e^{\lambda X}]$ is the Moment Generating Function (MGF)). Because, above holds for any λ , we find the λ that gives tightest bound.

$$\begin{aligned} \Pr[X \geq a] &\leq \exp(-(\lambda a - \psi_X(\lambda))) \\ &\leq \exp(-\sup_{\lambda} (\lambda a - \psi_X(\lambda))) \end{aligned}$$

Define $\psi_X^*(a) = \sup_{\lambda} (\lambda a - \ln(\mathbb{E}[e^{\lambda X}]))$ as the Cramer transform of X . Then,

$$\Pr[X \geq a] \leq e^{-\psi_X^*(a)}$$
□

Example. Chernoff bound for gaussian distribution. Let $X \sim \mathcal{G}(\mu, \sigma^2)$. Using Gaussian integral, MGF, $\mathbb{E}[e^{\lambda X}] = \exp(\lambda^2 \sigma^2/2 + \mu \lambda)$. Thus, $\psi_X(\lambda) = \mu \lambda + \sigma^2 \lambda^2/2$. Minimizing $\lambda a - \psi_X(\lambda)$ over λ , we get $\psi_X^*(a) = (\mu - a)^2/2\sigma^2$. Thus,

$$\Pr[X \geq a] \leq e^{-(\mu - a)^2/2\sigma^2}$$